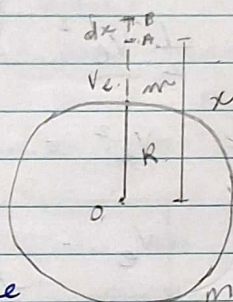


Escape velocity ( $V_e$ )

It is defined as the minimum velocity with which a body is to be projected vertically upwards from the surface of earth so that it leaves the earth's gravitational field completely and never returns back.

Maximum force to be applied on body at point A

$$F = \frac{GMm}{x^2}$$



Small work done in moving the body from point A to point B.

$$dW = F dx \cos 0^\circ$$

$$dW = \frac{GMm}{x^2} dx$$

Total work done in moving the body from the surface of earth to infinity.

$$\int dW = \int_R^\infty \frac{GMm}{x^2} dx$$

$$W = GMm \int_R^\infty x^{-2} dx$$

$$W = GMm \left[ \frac{x^{-2+1}}{-2+1} \right]_R^\infty = -GMm \left( \frac{1}{x} \right)_R^\infty$$

$$W = -GMm \left( \frac{1}{\infty} - \frac{1}{R} \right) = -GMm \left( 0 - \frac{1}{R} \right)$$

$$W = - \left( -\frac{GMm}{R} \right) = \frac{GMm}{R}$$

According to work energy theorem, this work done is equal to kinetic energy of the body.  
 i.e; work done = kinetic energy.

$$W = KE$$

$$\frac{GMm}{R} = \frac{1}{2}mv_e^2$$

$$\frac{1}{2}v_e^2 = \frac{GM}{R}$$

$$v_e^2 = \frac{2GM}{R}$$

$$v_e = \sqrt{\frac{2GM}{R}}$$

clearly escape velocity on the surface of a planet depends on mass of the planet and radius of the planet and is independent of mass of the body.

$$v_e = \sqrt{\frac{2GM}{R^2}}$$

$$v_e = \sqrt{2gR}$$

For earth,

$$g = 9.8 \text{ m/s}^2$$

$$R = 64 \times 10^5 \text{ m}$$

$$v_e = \sqrt{2 \times 9.8 \times 64 \times 10^5}$$

$$v_e = 11.2 \times 10^3 \text{ m/s}$$

$$v_e = 11.2 \text{ km/s}$$

Orbital velocity of a satellite.

A satellite is the name given to a small body which revolves around a comparatively larger body.

e.g; electron is the satellite of nucleus.

moon is the satellite of earth.

earth is the satellite of sun.

The first artificial satellite was launched by Russia namely Sputnik I and the first satellite launched by India was Aryabhata.

orbital velocity of a satellite is defined as the velocity with which it moves in its orbit around the earth.

Force on the satellite.

$$F = \frac{GMm}{r^2} \quad \text{--- (1)}$$

Also, this force acts as centripetal force.

$$F = \frac{mv_0^2}{r} \quad \text{--- (2)}$$

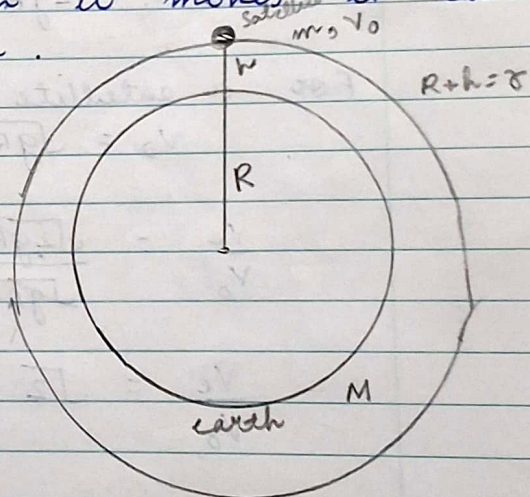
From (1) and (2).

$$\frac{GMm}{r^2} = \frac{mv_0^2}{r}$$

$$v_0^2 = \frac{GM}{r}$$

$$v_0 = \sqrt{\frac{GM}{r}}$$

For a satellite orbiting close to earth  
 $h \ll R$



$$V_o = \sqrt{\frac{GM}{R+h}}$$

$$V_o = \sqrt{\frac{GM}{R}}$$

$$V_o = \sqrt{\frac{GM}{R}} = \sqrt{gR} \quad \downarrow \quad 9.8 \text{ m/s}^2$$

Relation between escape velocity ( $V_e$ ) and orbital velocity ( $V_o$ ).  
we have,

$$V_e = \sqrt{2gR}$$

For a satellite orbiting close to earth,  
 $V_o = \sqrt{gR}$

$$\frac{V_e}{V_o} = \frac{\sqrt{2gR}}{\sqrt{gR}}$$

$$\frac{V_e}{V_o} = \sqrt{2}$$

$$V_e = \sqrt{2} V_o$$

Q. The orbital velocity of a satellite orbiting close to a planet is  $6 \text{ km/s}$ .

Find the escape velocity on its surface.

$$V_e = \sqrt{2} V_o = \sqrt{2} \times 6 = 8.48 \text{ km/s}$$

Q. what is infinity potential

Q. what is the gravitational potential of earth at infinity and what will be the gravitational potential energy of a body placed at infinity?

we have,

$$V = - \frac{GM}{r}$$

$$r = \infty.$$

$$\therefore V = - \frac{GM}{\infty} = 0.$$

$$\text{Also } U = V \cdot m.$$

$$U = 0 \cdot m = 0.$$